

Game Theory-Assignment5

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Exercise 11

Let $A \in \mathbb{R}^{n \times m}$ define a zero-sum matrix game.

1. Adding a constant to all entries

Let $B = A + \lambda \mathbf{1}$, where $\lambda \in \mathbb{R}$ and every entry of A is increased by λ .

For mixed strategies $y \in \Delta_n$ and $z \in \Delta_m$, the expected payoff in game A is

$$u_A(y, z) = y^T A z.$$

In game B :

$$u_B(y, z) = y^T (A + \lambda \mathbf{1}) z = y^T A z + \lambda y^T \mathbf{1} z.$$

Since y, z are probability vectors,

$$y^T \mathbf{1} z = 1.$$

Thus

$$u_B(y, z) = u_A(y, z) + \lambda.$$

Hence

$$\max_y \min_z u_B(y, z) = \max_y \min_z (u_A(y, z) + \lambda) = \max_y \min_z u_A(y, z) + \lambda.$$

Similarly

$$\min_z \max_y u_B(y, z) = \min_z \max_y u_A(y, z) + \lambda.$$

Therefore the optimal strategies are unchanged. A pair (y^*, z^*) is a mixed saddle-point equilibrium for A iff it is for B .

2. Monotone increasing transformation (pure equilibrium preserved)

Let C be defined by

$$C_{ij} = f(A_{ij}),$$

where f is monotone increasing:

$$x \leq y \Rightarrow f(x) \leq f(y).$$

Suppose (i^*, j^*) is a pure saddle point of A :

$$A_{i^*j} \leq A_{i^*j^*} \leq A_{ij^*} \quad \forall i, j.$$

Apply monotonicity:

$$f(A_{i^*j}) \leq f(A_{i^*j^*}) \leq f(A_{ij^*}).$$

Thus

$$C_{i^*j} \leq C_{i^*j^*} \leq C_{ij^*}.$$

Hence (i^*, j^*) is also a pure saddle point of C .

3. Strictly increasing transformation (iff)

Assume f strictly increasing:

$$x < y \Rightarrow f(x) < f(y).$$

From part 2: pure saddle in A pure saddle in C .

Conversely suppose (i^*, j^*) is saddle in C :

$$f(A_{i^*j}) \leq f(A_{i^*j^*}) \leq f(A_{ij^*}).$$

Since f strictly increasing, it preserves order reversibly:

$$A_{i^*j} \leq A_{i^*j^*} \leq A_{ij^*}.$$

Thus (i^*, j^*) saddle in A .

4. Affine strictly increasing transform (mixed equilibria)

Let

$$f(x) = \alpha x + \lambda, \quad \alpha > 0.$$

Then

$$C = \alpha A + \lambda \mathbf{1}.$$

Expected payoff:

$$u_C(y, z) = \alpha y^T A z + \lambda.$$

Hence

$$\max_y \min_z u_C(y, z) = \alpha \max_y \min_z y^T A z + \lambda.$$

Since $\alpha > 0$, scaling does not change argmax or argmin.

Thus optimal mixed strategies unchanged.

Exercise 12

Consider a bimatrix game. For player P_i :

Pure security level:

$$v^{\text{pure}} = \max_{s_i \in S_i} \min_{s_{-i}} u_i(s_i, s_{-i}).$$

Mixed security level:

$$v^{\text{mixed}} = \max_{\sigma_i \in \Delta(S_i)} \min_{\sigma_{-i}} u_i(\sigma_i, \sigma_{-i}).$$

Pure strategies are special cases of mixed strategies (degenerate distributions).

Thus the feasible set of mixed strategies strictly contains pure ones.

Therefore

$$\max_{\sigma_i} \min_{\sigma_{-i}} u_i \geq \max_{s_i} \min_{s_{-i}} u_i.$$

So

$$v^{\text{mixed}} \geq v^{\text{pure}}.$$

Meaning the mixed security level is always at least as good for the player.